

**15.3** Solution: Similar to Problem 15.2, we have

$$\frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} = \frac{d^2N}{d(\hbar\omega)d\Omega} \cdot \frac{d\sigma}{d\Omega_p}.$$

For classical elastic hard sphere scattering, the cross section is

$$\frac{d\sigma}{d\Omega_p} = \frac{1}{4}R^2,$$

and the differential number spectrum per unit energy interval and per unit solid angle of soft photons, including the relativistic effect, is given by Eq. (15.3),

$$\frac{d^2N}{d(\hbar\omega)d\Omega} = \frac{e^2}{4\hbar\pi^2c} \frac{1}{\hbar\omega} \sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i^* \cdot \left( \frac{\boldsymbol{\beta}_f}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_f} - \frac{\boldsymbol{\beta}_i}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_i} \right) \right|^2.$$

For the initial velocity part, we have

$$\boldsymbol{\beta}_i = \beta(0, 0, 1), \quad \boldsymbol{\beta}_f = \beta(\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p),$$

and

$$\mathbf{n} = (\sin \theta, 0, \cos \theta), \quad \boldsymbol{\epsilon}_1 = (\cos \theta, 0, -\sin \theta), \quad \boldsymbol{\epsilon}_2 = (0, 1, 0).$$

Then,

$$\frac{\boldsymbol{\epsilon}_1^* \cdot \boldsymbol{\beta}_i}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_i} = -\frac{\beta \sin \theta}{1 - \beta \cos \theta}, \quad \frac{\boldsymbol{\epsilon}_2^* \cdot \boldsymbol{\beta}_i}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_i} = 0.$$

For the final velocity part, we cannot calculate the necessary ingredients in the same way. Instead, we can rotate the coordinate system around the  $y$ -axis by an angle of  $\theta$ , assume that  $\mathbf{n}$  is along the  $z$ -axis and the final velocity is along the  $(\theta_p, \phi_p)$  direction in spherical coordinates. Following this configuration, we have

$$\boldsymbol{\beta}_f = \beta(\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p), \quad \mathbf{n} = (0, 0, 1), \quad \boldsymbol{\epsilon}_1 = (1, 0, 0), \quad \boldsymbol{\epsilon}_2 = (0, 1, 0).$$

Then,

$$\frac{\boldsymbol{\epsilon}_1^* \cdot \boldsymbol{\beta}_f}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_f} = \frac{\beta \sin \theta_p \cos \phi_p}{1 - \beta \cos \theta_p}, \quad \frac{\boldsymbol{\epsilon}_2^* \cdot \boldsymbol{\beta}_f}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_f} = \frac{\beta \sin \theta_p \sin \phi_p}{1 - \beta \cos \theta_p}.$$

Now we can write the differential cross section as

$$\begin{aligned} \frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} &= \frac{e^2}{4\hbar\pi^2c} \frac{\beta^2}{\hbar\omega} \frac{R^2}{4} \left[ \left( \frac{\sin \theta_p \cos \phi_p}{1 - \beta \cos \theta_p} + \frac{\sin \theta}{1 - \beta \cos \theta} \right)^2 + \frac{\sin^2 \theta_p \sin^2 \phi_p}{(1 - \beta \cos \theta_p)^2} \right] \\ &= \frac{e^2}{4\hbar\pi^2c} \frac{\beta^2}{\hbar\omega} \frac{R^2}{4} \left[ \frac{2 \sin \theta \sin \theta_p \cos \phi_p}{(1 - \beta \cos \theta)(1 - \beta \cos \theta_p)} + \frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} + \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^2} \right]. \end{aligned}$$

Upon integrating with respect to  $\Omega_p$ , the first term is clearly 0 and last term will be multiplied by  $4\pi$ . The second term can be easily evaluated,

$$\begin{aligned} \int \frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} d\Omega_p &= 2\pi \int_0^\pi \frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} \sin \theta_p d\theta_p = 2\pi \int_{-1}^1 \frac{1 - x^2}{(1 - \beta x)^2} dx \\ &= \frac{2\pi}{\beta} \int_{-1}^1 (1 - x^2) d \left( \frac{1}{1 - \beta x} \right) = \frac{4\pi}{\beta} \int_{-1}^1 \frac{x}{1 - \beta x} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{4\pi}{\beta} \int_{-1}^1 \left( -\frac{1}{\beta} + \frac{1}{\beta(1-\beta x)} \right) dx = \frac{4\pi}{\beta} \left( -\frac{2}{\beta} - \frac{2}{\beta^2} \log(1-\beta x) \Big|_{-1}^1 \right) \\
&= 4\pi \left( \frac{1}{\beta^3} \log \left( \frac{1+\beta}{1-\beta} \right) - \frac{2}{\beta^2} \right).
\end{aligned}$$

Finally,

$$\begin{aligned}
\frac{d^2\sigma}{d(\hbar\omega)d\Omega} &= \int \frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} d\Omega_p \\
&= \frac{e^2}{4\hbar\pi^2 c} \frac{\beta^2}{\hbar\omega} \frac{R^2}{4} \cdot 4\pi \cdot \left[ \frac{\sin^2 \theta}{(1-\beta \cos \theta)^2} + \frac{1}{\beta^3} \log \left( \frac{1+\beta}{1-\beta} \right) - \frac{2}{\beta^2} \right] \\
&= \frac{R^2}{4\pi} \frac{e^2}{\hbar c} \frac{\beta^2}{\hbar\omega} \left[ \frac{\sin^2 \theta}{(1-\beta \cos \theta)^2} + \frac{1}{\beta^3} \log \left( \frac{1+\beta}{1-\beta} \right) - \frac{2}{\beta^2} \right].
\end{aligned}$$

In the nonrelativistic limit,  $\beta \rightarrow 0$ , the logarithm function can be expanded as

$$\log \left( \frac{1+\beta}{1-\beta} \right) = 2\beta + \frac{2}{3}\beta^3 + o(\beta^3),$$

and the corresponding limiting value of the differential cross section will reduce to that of Problem 15.2.